Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

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8 Proofs Involving Sets

Use the methods introduced in this chapter to prove the following statements.

2. Prove that $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$.

Proof. First we show $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. Suppose $a \in \{6n : n \in \mathbb{Z}\}$. This means $a = 6n$ for some integer n. Therefore $a = 2(3n)$ and $a = 3(2n)$. From $a = 2(3n)$, it follows that a is a multiple of 2, so $a \in \{2n : n \in \mathbb{Z}\}$. From $a = 3(2n)$, it follows that a is a multiple of 3, so $a \in \{3n : n \in \mathbb{Z}\}$. Thus by definition of intersection of two sets, we have $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. Therefore $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$.

Next we show $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$.
Suppose $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. By definition of intersection of two sets, this means $a \in \{2n : n \in \mathbb{Z}\}$ and $a \in \{3n : n \in \mathbb{Z}\}$. Since $a \in \{2n : n \in \mathbb{Z}\}$, we know $a = 2k$ for some $k \in \mathbb{Z}$. Thus a is even. Since $a \in \{3n : n \in \mathbb{Z}\}$, we know $a = 3l$ for some $l \in \mathbb{Z}$. Since 3 is odd, if l were odd then $3l$ would be odd; as $a = 3l$ is even, it follows that l is even. Then $l = 2j$ for some $j \in \mathbb{Z}$. So we have $a = 3l = 3(2j) = 6j$. Since $j \in \mathbb{Z}$, it follows that $a \in \{6n : n \in \mathbb{Z}\}$. Thus $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$.

Therefore, since $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ and $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$, it follows that $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. □

4. If $m, n \in \mathbb{Z}$, then $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$.

Proof. Suppose $a \in \{x \in \mathbb{Z} : mn \mid x\}$. Then $a = mnk$ for some $k \in \mathbb{Z}$. So we have $a = m(nk)$ and $a = n(mk)$. By definition of divisibility, $m \mid a$ and $n \mid a$. Thus $a \in \{x \in \mathbb{Z} : m \mid x\}$ and $a \in \{x \in \mathbb{Z} : n \mid x\}$. By definition of intersection of two sets, we have $a \in \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$. Therefore $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$. □

6. Suppose A, B and C are sets. Prove that if $A \subseteq B$, then $A - C \subseteq B - C$.

Proof. We prove the contrapositive: if $A - C \nsubseteq B - C$, then $A \nsubseteq B$. Assume $A - C \nsubseteq B - C$. Then there exists an element $x \in A - C$ such that $x \notin B - C$. Since $x \in A - C$, we have $x \in A$ and $x \notin C$. Since $x \notin B - C$, it follows that $x \notin B$ or $x \in C$. Because $x \notin C$, it follows $x \notin B$. Since $x \in A$ and $x \notin B$, by definition of a subset, we have $A \nsubseteq B$. □

8. If A, B and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Proof. Observe the following sequence of equalities.

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\begin{aligned}
A \cup (B \cap C) &= \{x : x \in A \vee x \in B \cap C\} && \text{(definition of } \cup) \\
&= \{x : x \in A \vee (x \in B \wedge x \in C)\} && \text{(definition of } \cap) \\
&= \{x : (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} && \text{(distributive law)} \\
&= \{x : x \in A \cup B \wedge x \in A \cup C\} && \text{(definition of } \cup) \\
&= \{x : x \in (A \cup B) \cap (A \cup C)\} && \text{(definition of } \cap) \\
&= (A \cup B) \cap (A \cup C) && \text{(definition of set-builder)}
\end{aligned}
$$

□